

Time-independent perturbation theory[⊛]

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Many problems in quantum mechanics need to be solved using approximation methods. In many situations, these methods are sufficient to obtain the energy spectrum of the system, as well as to understand the corresponding physical phenomena.

Objective This, therefore, is the objective of this chapter: describe the time-independent perturbation theory, followed by a discussion of some applications. More specifically, we will explore the van Vleck susceptibility and the Stark effect, and use perturbative techniques as a tool.

[⊛]This book has a companion website hosting complementary materials. Visit this URL to access it: <https://www.elsevier.com/books-and-journals/book-companion/9780443328268>.

Throughout this chapter, we will consider two cases: the first where there are no degenerate energy states, and the second where degenerate energy states are present.

8.1 Non-degenerate case

Consider a non-perturbed Hamiltonian $\hat{\mathcal{H}}_0$, whose vectors are given by $\{|n^{(0)}\rangle\}$:

$$\hat{\mathcal{H}}_0|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle, \quad (8.1)$$

being $E_n^{(0)}$ the corresponding eigenvalues. We are assuming that the vectors of this basis are orthogonal, that is:

$$\langle m^{(0)}|n^{(0)}\rangle = \delta_{mn}. \quad (8.2)$$

Consider now a perturbation $\hat{\mathcal{W}}$, where the strength of this perturbation is controlled by the parameter λ , which ranges between 0 and 1. In this way, the total Hamiltonian can be written as:

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \lambda\hat{\mathcal{W}}. \quad (8.3)$$

Some important observations: (i) the perturbation $\hat{\mathcal{W}}$ is a physical observable and, therefore, must be a Hermitian operator; (ii) $\hat{\mathcal{W}}$ must be expressed in the basis vectors $\{|n^{(0)}\rangle\}$. Clearly, $\hat{\mathcal{W}}$ is non-diagonal, because if it were, $\hat{\mathcal{H}}$ would also be diagonal. Thus, we can assert that there exists another basis vectors $\{|n\rangle\}$ in which $\hat{\mathcal{H}}$ is diagonal. Thus:

$$\hat{\mathcal{H}}|n\rangle = E_n|n\rangle, \quad (8.4)$$

where E_n are the eigenvalues of the total Hamiltonian. Similarly to the non-perturbed case, the vectors for $\hat{\mathcal{H}}$ are orthogonal:

$$\langle m|n\rangle = \delta_{mn}. \quad (8.5)$$

Objective The objective of this section is, therefore, to approximately obtain the eigenvalues E_n and vectors $|n\rangle$ of the total Hamiltonian.

For this purpose, we will consider the eigenvalues E_n and the vectors $|n\rangle$ as an expansion with respect to the parameter λ , as follows:

$$|n\rangle = |n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \lambda^3|n^{(3)}\rangle + \dots \quad (8.6)$$

and

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \lambda^3 E_n^{(3)} + \dots \quad (8.7)$$

By substituting the two equations above into the total Hamiltonian (Eq. (8.4)), we obtain:

$$\begin{aligned} & \left[\hat{\mathcal{H}}_0 + \lambda \hat{\mathcal{V}} \right] \left[|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \lambda^3 |n^{(3)}\rangle \right] = \\ & \left[E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \lambda^3 E_n^{(3)} \right] \left[|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \lambda^3 |n^{(3)}\rangle \right]. \end{aligned} \quad (8.8)$$

We can compare terms of the same order in λ , resulting in:

$$\lambda^0: \quad \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(0)}\rangle = 0 \quad (8.9)$$

$$\lambda^1: \quad \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(1)}\rangle = -\hat{\mathcal{V}}|n^{(0)}\rangle + E_n^{(1)}|n^{(0)}\rangle \quad (8.10)$$

$$\lambda^2: \quad \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(2)}\rangle = -\hat{\mathcal{V}}|n^{(1)}\rangle + E_n^{(1)}|n^{(1)}\rangle + E_n^{(2)}|n^{(0)}\rangle \quad (8.11)$$

$$\lambda^3: \quad \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(3)}\rangle = -\hat{\mathcal{V}}|n^{(2)}\rangle + E_n^{(1)}|n^{(2)}\rangle + E_n^{(2)}|n^{(1)}\rangle + E_n^{(3)}|n^{(0)}\rangle. \quad (8.12)$$

It is left as an exercise for the reader to derive a recurrence relation for the equations above.

The orthogonality of the vectors $\{|n\rangle\}$ also provides an equation in powers of λ , given by (Eq. (8.6) into Eq. (8.5)):

$$\begin{aligned} \langle n|n\rangle &= 1 = 1 + \lambda 0 + \lambda^2 0 + \lambda^3 0 \\ &= \left[\langle n^{(0)}| + \lambda \langle n^{(1)}| + \lambda^2 \langle n^{(2)}| + \lambda^3 \langle n^{(3)}| \right] \\ &\quad \times \left[|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \lambda^3 |n^{(3)}\rangle \right]. \end{aligned} \quad (8.13)$$

Comparing terms of the same order in λ , we obtain:

$$\lambda^0: \quad \langle n^{(0)}|n^{(0)}\rangle = 1 \quad (8.14)$$

$$\lambda^1: \quad \langle n^{(0)}|n^{(1)}\rangle + \langle n^{(1)}|n^{(0)}\rangle = 0 \quad (8.15)$$

$$\lambda^2: \quad \langle n^{(0)}|n^{(2)}\rangle + \langle n^{(1)}|n^{(1)}\rangle + \langle n^{(2)}|n^{(0)}\rangle = 0 \quad (8.16)$$

$$\lambda^3: \quad \langle n^{(0)}|n^{(3)}\rangle + \langle n^{(1)}|n^{(2)}\rangle + \langle n^{(2)}|n^{(1)}\rangle + \langle n^{(3)}|n^{(0)}\rangle = 0. \quad (8.17)$$

It is also possible to obtain a recurrence relation for the equations above, which is left as an exercise for the reader.

In the present non-degenerate case, each vector $|n^{(0)}\rangle$ is associated with only one energy eigenvalue $E_n^{(0)}$. In the following subsections, we will derive the first and second order corrections for the energy ($E_n^{(1)}$ and $E_n^{(2)}$), and for the vectors ($|n^{(1)}\rangle$ and $|n^{(2)}\rangle$).

■ Example 8.1

Let us consider a non-perturbed Hamiltonian $\hat{\mathcal{H}}_0$ and a perturbation $\hat{\mathcal{W}}$ described by:

$$\hat{\mathcal{H}}_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{pmatrix} \quad \text{and} \quad \hat{\mathcal{W}} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (8.18)$$

It is important to emphasize that the purpose of this example is merely to demonstrate the use of the formalism developed above. Thus, the non-perturbed Hamiltonian $\hat{\mathcal{H}}_0$ and the perturbation $\hat{\mathcal{W}}$ have arbitrary values without physical significance. In the Applications section of this chapter, real examples of existing physical situations will be described. Also, note that $\hat{\mathcal{H}}_0$ is diagonal, and $\hat{\mathcal{W}}$ is not, as expected. If $\hat{\mathcal{H}}_0$ were not diagonal, the studied formalism could not be applied, as Eq. (8.1) requires the non-perturbed Hamiltonian to be diagonal in the basis vectors $\{|n^{(0)}\rangle\}$. If the perturbation were also diagonal, we would not need this theory, because the total Hamiltonian $\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \lambda \hat{\mathcal{W}}$ would also be diagonal.

Thus, we can write the eigenvalues for the non-perturbed system as:

$$E_1^{(0)} = 1, \quad E_2^{(0)} = 3 \quad \text{and} \quad E_3^{(0)} = -2, \quad (8.19)$$

as well as its vectors:

$$|1^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |2^{(0)}\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |3^{(0)}\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (8.20)$$

This system proposed in Eq. (8.18) is quite simple and, therefore, has an exact analytical solution. Thus, we will first obtain the exact solution and then take the limit $\lambda \rightarrow 0$. Throughout this chapter, as we develop the formalism of Perturbation Theory, we will keep returning to this example, as well as comparing it with the analytical solution obtained below.

Analytical solution: Consider the eigenvalue equation $\hat{\mathcal{H}}|n\rangle = E_n|n\rangle$ to obtain the eigenvalues E_n and the corresponding vectors $|n\rangle$. Although straightforward, this task is somewhat time-consuming and space-demanding, and since it lies outside the objectives of this example, we will present only the final result,¹ along with its expansions for the case $\lambda \rightarrow 0$. Thus, the eigenvalues are:

$$E_1 = 2 - \sqrt{\lambda^2 + 1} \xrightarrow{\lambda \rightarrow 0} 1 - \frac{\lambda^2}{2}, \quad (8.21)$$

¹Solution obtained using a software [19].

$$E_2 = 2 + \sqrt{\lambda^2 + 1} \xrightarrow{\lambda \rightarrow 0} 3 + \frac{\lambda^2}{2}, \quad (8.22)$$

$$E_3 = -2 + \lambda \xrightarrow{\lambda \rightarrow 0} -2 + \lambda. \quad (8.23)$$

The vector for $n = 1$ is given by:

$$|1\rangle = \begin{pmatrix} a_+ \\ b_+ \\ 0 \end{pmatrix} = a_+ |1^{(0)}\rangle + b_+ |2^{(0)}\rangle, \quad (8.24)$$

where

$$a_{\pm} = \frac{1}{\sqrt{2}} \frac{(\sqrt{\lambda^2 + 1} \pm 1)}{[(\lambda^2 + 1) \pm \sqrt{\lambda^2 + 1}]^{1/2}} \xrightarrow{\lambda \rightarrow 0} \begin{cases} a_+ \approx 1 - \frac{\lambda^2}{8} \\ a_- \approx \frac{\lambda}{2} \end{cases} \quad (8.25)$$

and

$$b_{\pm} = \mp \frac{1}{\sqrt{2}} \frac{\lambda}{[(\lambda^2 + 1) \pm \sqrt{\lambda^2 + 1}]^{1/2}} \xrightarrow{\lambda \rightarrow 0} \begin{cases} b_+ \approx -\frac{\lambda}{2} \\ b_- \approx 1 - \frac{\lambda^2}{8} \end{cases} \quad (8.26)$$

Thus, the vector for $n = 1$ can be written as:

$$|1\rangle \xrightarrow{\lambda \rightarrow 0} \left(1 - \frac{\lambda^2}{8}\right) |1^{(0)}\rangle - \frac{\lambda}{2} |2^{(0)}\rangle; \quad (8.27)$$

which can be organized in powers of λ :

$$|1\rangle \rightarrow |1^{(0)}\rangle - \frac{\lambda}{2} |2^{(0)}\rangle - \frac{\lambda^2}{8} |1^{(0)}\rangle. \quad (8.28)$$

For $n = 2$ the vector reads as:

$$|2\rangle = \begin{pmatrix} a_- \\ b_- \\ 0 \end{pmatrix} = a_- |1^{(0)}\rangle + b_- |2^{(0)}\rangle$$

$$\xrightarrow{\lambda \rightarrow 0} \frac{\lambda}{2} |1^{(0)}\rangle + \left(1 - \frac{\lambda^2}{8}\right) |2^{(0)}\rangle; \quad (8.29)$$

which can also be organized in powers of λ :

$$|2\rangle \rightarrow |2^{(0)}\rangle + \frac{\lambda}{2} |1^{(0)}\rangle - \frac{\lambda^2}{8} |2^{(0)}\rangle. \quad (8.30)$$

Finally, for $n = 3$:

$$|3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = |3^{(0)}\rangle. \quad (8.31)$$

■

8.1.1 First order correction to energy and vectors

To obtain the first order energy correction $E_n^{(1)}$ to the energy E_n , we will apply the *bra* $\langle n^{(0)}|$, from the left, to Eq. (8.10):

$$\langle n^{(0)}|(\hat{\mathcal{H}}_0 - E_n^{(0)})|n^{(1)}\rangle = -\langle n^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle + E_n^{(1)}\langle n^{(0)}|n^{(0)}\rangle. \quad (8.32)$$

We know that $\langle n^{(0)}|\hat{\mathcal{H}}_0 = \langle n^{(0)}|E_n^{(0)}$ and therefore the left-hand side of the above equation is zero. Considering $\langle n^{(0)}|n^{(0)}\rangle = 1$, we obtain the first order energy correction:

$$E_n^{(1)} = \langle n^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle \quad (8.33)$$

To think What does the mathematical object on the right-hand side of the above equation represent? In other words, how can we obtain the first energy correction?

As discussed throughout this book (see, for example, Section 3.3), mathematical objects such as $\langle n^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle$ represent matrix elements. This particular one represents the n -th row and n -th column (diagonal elements) of the perturbative matrix $\hat{\mathcal{W}}$ written in the basis of the unperturbed Hamiltonian $\hat{\mathcal{H}}_0$.

As for the first correction to the vectors, this development is somewhat more elaborate, and for this reason, it is presented separately in Box 8.1 – we invite the reader to review the corresponding calculations. As a result, we obtain:

$$|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle \quad (8.34)$$

Note, therefore, that the first correction $|n^{(1)}\rangle$ to the state $|n\rangle$ depends on all the unperturbed states $|m^{(0)}\rangle$. Moreover, the weight of this dependence is given by the element $\langle m^{(0)}|n^{(1)}\rangle$ (Eq. (8.39)), which becomes larger as the energy gap between the states $|m^{(0)}\rangle$ and $|n^{(0)}\rangle$ decreases. In other words,

there is a mixing of states, where the states $|m^{(0)}\rangle$ that are energetically closer to $|n^{(0)}\rangle$ contribute more significantly to this mixture.

■ Example 8.2

Considering the Hamiltonian and perturbation from Example 8.1, we can calculate the first order energy correction using Eq. (8.33), which consists only of the diagonal elements of the perturbation:

$$E_1^{(1)} = 0, \quad E_2^{(1)} = 0 \quad \text{and} \quad E_3^{(1)} = 1.$$

As for the vectors, the first order correction can be obtained using Eq. (8.34). Thus:

$$|1^{(1)}\rangle = \frac{\mathcal{W}_{21}}{E_{12}}|2^{(0)}\rangle + \frac{\mathcal{W}_{31}}{E_{13}}|3^{(0)}\rangle = -\frac{1}{2}|2^{(0)}\rangle, \quad (8.35)$$

$$|2^{(1)}\rangle = \frac{\mathcal{W}_{12}}{E_{21}}|1^{(0)}\rangle + \frac{\mathcal{W}_{32}}{E_{23}}|3^{(0)}\rangle = \frac{1}{2}|1^{(0)}\rangle, \quad (8.36)$$

$$|3^{(1)}\rangle = \frac{\mathcal{W}_{13}}{E_{31}}|1^{(0)}\rangle + \frac{\mathcal{W}_{23}}{E_{32}}|2^{(0)}\rangle = 0, \quad (8.37)$$

where $\mathcal{W}_{ab} = \langle a^{(0)}|\hat{\mathcal{W}}|b^{(0)}\rangle$ and $E_{ab} = E_a^{(0)} - E_b^{(0)}$. ■

Box 8.1 Detailed description for the first order correction $|n^{(1)}\rangle$ to the vector $|n\rangle$

To obtain the first order correction $|n^{(1)}\rangle$ to the vector $|n\rangle$, we apply the bra $\langle m^{(0)}|$ from the left to Eq. (8.10):

$$\langle m^{(0)}|(\hat{\mathcal{H}}_0 - E_n^{(0)})|n^{(1)}\rangle = -\langle m^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle + E_n^{(1)}\langle m^{(0)}|n^{(0)}\rangle. \quad (8.38)$$

The vectors of $\hat{\mathcal{H}}_0$ are orthogonal (see Eq. (8.2)) and therefore the above equation can be rewritten as:

$$\langle m^{(0)}|n^{(1)}\rangle = \frac{\langle m^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}. \quad (8.39)$$

Now, let's apply the ket $|m^{(0)}\rangle$ on both sides of the above equation, from the left, and sum over all states m , except for n :

$$\sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)}|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle. \quad (8.40)$$

The left-hand side of the above equation needs to be carefully analyzed:

$$\begin{aligned}
 & \sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(1)}\rangle \\
 &= \sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(1)}\rangle + |n^{(0)}\rangle \langle n^{(0)} | n^{(1)}\rangle - |n^{(0)}\rangle \langle n^{(0)} | n^{(1)}\rangle \quad (8.41) \\
 &= \left(\sum_m |m^{(0)}\rangle \langle m^{(0)} | \right) |n^{(1)}\rangle - |n^{(0)}\rangle \langle n^{(0)} | n^{(1)}\rangle \\
 &= |n^{(1)}\rangle - |n^{(0)}\rangle \langle n^{(0)} | n^{(1)}\rangle.
 \end{aligned}$$

Above, the completeness relation was used: $\sum_m |m^{(0)}\rangle \langle m^{(0)}| = \mathbb{1}$. We need to determine the inner product $\langle n^{(0)} | n^{(1)}\rangle$. For this purpose, let's assume, without loss of generality, that the numbers $\langle n^{(i)} | n^{(j)}\rangle$ are real, which implies:

$$\langle n^{(i)} | n^{(j)}\rangle = \langle n^{(j)} | n^{(i)}\rangle. \quad (8.42)$$

Considering Eq. (8.15), we have:

$$\langle n^{(0)} | n^{(1)}\rangle = \langle n^{(1)} | n^{(0)}\rangle = 0. \quad (8.43)$$

Finally, the left-hand side of Eq. (8.40) reads as:

$$\sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(1)}\rangle = |n^{(1)}\rangle. \quad (8.44)$$

By substituting Eq. (8.44) into Eq. (8.40), we find the first order correction for the vectors:

$$|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m^{(0)} | \hat{\mathcal{V}} | n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle. \quad (8.45)$$

To think Why do we sum over all states except for n in Eq. (8.40)?

8.1.2 Second order correction to energy and vectors

Analogous to the previous case, to obtain the second order correction $E_n^{(2)}$ to the energy E_n , we apply the *bra* $\langle n^{(0)}|$, from the left, to Eq. (8.11):

$$\langle n^{(0)} | \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) | n^{(2)}\rangle = -\langle n^{(0)} | \hat{\mathcal{V}} | n^{(1)}\rangle + E_n^{(1)} \langle n^{(0)} | n^{(1)}\rangle + E_n^{(2)} \langle n^{(0)} | n^{(0)}\rangle. \quad (8.46)$$

The term on the left-hand side of the above equation is zero, since $\langle n^{(0)} | \hat{\mathcal{H}}_0 = \langle n^{(0)} | E_n^{(0)}$. Additionally, $\langle n^{(0)} | n^{(0)} \rangle = 1$ (Eq. (8.2)) and $\langle n^{(0)} | n^{(1)} \rangle = 0$ (Eq. (8.43)). Thus:

$$E_n^{(2)} = \langle n^{(0)} | \hat{\mathcal{W}} | n^{(1)} \rangle. \quad (8.47)$$

By substituting Eq. (8.34) into Eq. (8.47), we obtain the desired result, which is the second order energy correction:

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} \quad (8.48)$$

where we have considered $\langle n^{(0)} | \hat{\mathcal{W}} | m^{(0)} \rangle \langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle = \langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle^* \langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle = |\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle|^2$, since $\hat{\mathcal{W}}$, being a perturbation to the Hamiltonian, must be an observable and, therefore, Hermitian.

To think What does the mathematical object in the numerator of Eq. (8.48) represent?

As discussed for the first order energy correction, mathematical objects such as $\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle$ represent a matrix element of the perturbation $\hat{\mathcal{W}}$. Specifically, this represents the m -th row and n -th column of the perturbative matrix $\hat{\mathcal{W}}$, written in the basis of the unperturbed Hamiltonian $\hat{\mathcal{H}}_0$. Therefore, these are the elements *off* the main diagonal. This energy correction is larger when the energy gap between the states $|m^{(0)}\rangle$ and $|n^{(0)}\rangle$ is smaller.

As for the second order correction to the vectors, this development is presented separately in Box 8.2, similar to what we did for the first order correction to the vectors (Box 8.1). We invite the reader to review the corresponding calculations. As a result, we obtain:

$$|n^{(2)}\rangle = -\frac{1}{2} \sum_{m \neq n} \frac{|\mathcal{W}_{mn}|^2}{(E_{nm})^2} |n^{(0)}\rangle - \sum_{m \neq n} \frac{\mathcal{W}_{nn} \mathcal{W}_{mn}}{(E_{nm})^2} |m^{(0)}\rangle + \sum_{m \neq n} \sum_{l \neq n} \frac{\mathcal{W}_{ml} \mathcal{W}_{ln}}{E_{nm} E_{nl}} |m^{(0)}\rangle \quad (8.49)$$

where $\mathcal{W}_{ab} = \langle a^{(0)} | \hat{\mathcal{W}} | b^{(0)} \rangle$ and $E_{ab} = E_a^{(0)} - E_b^{(0)}$. Analogous to the first order correction to the vectors, there is also a mixing of states. That is, the correction $|n^{(2)}\rangle$ to the state $|n\rangle$ depends on other states $|m^{(0)}\rangle$. The closer (energetically) the state $|m^{(0)}\rangle$ is to the state $|n^{(0)}\rangle$, the greater this mixing is, due to the denominator of Eq. (8.49).

■ Example 8.3

It is now possible to calculate the second order energy correction

for the examples we are dealing with (Examples 8.1 and 8.2). For this purpose, consider the Hamiltonian and perturbation from Example 8.1, as well as Eq. (8.48). Thus:

$$E_1^{(2)} = \frac{|\mathcal{W}_{21}|^2}{E_{12}} + \frac{|\mathcal{W}_{31}|^2}{E_{13}} = \frac{1^2}{1-3} + \frac{0^2}{1-(-2)} = -\frac{1}{2}, \quad (8.50)$$

$$E_2^{(2)} = \frac{|\mathcal{W}_{12}|^2}{E_{21}} + \frac{|\mathcal{W}_{32}|^2}{E_{23}} = \frac{1^2}{3-1} + \frac{0^2}{3-(-2)} = \frac{1}{2}, \quad (8.51)$$

$$E_3^{(2)} = \frac{|\mathcal{W}_{13}|^2}{E_{31}} + \frac{|\mathcal{W}_{23}|^2}{E_{32}} = \frac{0^2}{-2-1} + \frac{0^2}{-2-3} = 0. \quad (8.52)$$

To obtain the second order correction in the vectors, we need to recall Eq. (8.49). For the case $n = 1$ it reads as:

$$\begin{aligned} |1^{(2)}\rangle = & -\frac{1}{2} \left(\frac{|\mathcal{W}_{21}|^2}{E_{12}} + \frac{|\mathcal{W}_{31}|^2}{E_{13}} \right) |1^{(0)}\rangle - \frac{\mathcal{W}_{11}\mathcal{W}_{21}}{E_{12}^2} |2^{(0)}\rangle - \frac{\mathcal{W}_{11}\mathcal{W}_{31}}{E_{13}^2} |3^{(0)}\rangle \\ & + \frac{\mathcal{W}_{22}\mathcal{W}_{21}}{E_{12}E_{12}} |2^{(0)}\rangle + \frac{\mathcal{W}_{23}\mathcal{W}_{31}}{E_{12}E_{13}} |2^{(0)}\rangle + \frac{\mathcal{W}_{32}\mathcal{W}_{21}}{E_{13}E_{12}} |3^{(0)}\rangle \\ & + \frac{\mathcal{W}_{33}\mathcal{W}_{31}}{E_{13}E_{13}} |3^{(0)}\rangle. \end{aligned} \quad (8.53)$$

Except for the elements $\mathcal{W}_{12} = \mathcal{W}_{21} = \mathcal{W}_{33} = 1$, all terms of the perturbation $\hat{\mathcal{W}}$ are zero. Thus, it is easy to verify that the above equation reads as:

$$|1^{(2)}\rangle = -\frac{1}{8} |1^{(0)}\rangle. \quad (8.54)$$

We leave it as an exercise for the reader to demonstrate that:

$$|2^{(2)}\rangle = -\frac{1}{8} |2^{(0)}\rangle \quad \text{and} \quad |3^{(2)}\rangle = 0. \quad (8.55)$$

■

■ Example 8.4

Considering the Hamiltonian and perturbation from Example 8.1, as well as the results from Examples 8.2 and 8.3, it is now possible to write the three energy eigenvalues:

$$E_1 = 1 - \frac{\lambda^2}{2}, \quad E_2 = 3 + \frac{\lambda^2}{2} \quad \text{and} \quad E_3 = -2 + \lambda; \quad (8.56)$$

as well as the three vectors for the perturbed Hamiltonian:

$$|1\rangle = |1^{(0)}\rangle - \frac{\lambda}{2}|2^{(0)}\rangle - \frac{\lambda^2}{8}|1^{(0)}\rangle, \quad (8.57)$$

$$|2\rangle = |2^{(0)}\rangle + \frac{\lambda}{2}|1^{(0)}\rangle - \frac{\lambda^2}{8}|2^{(0)}\rangle, \quad (8.58)$$

$$|3\rangle = |3^{(0)}\rangle. \quad (8.59)$$

It is important to verify that the results above, obtained through Perturbation Theory, recover those obtained through the exact analytical solution with the approximation $\lambda \rightarrow 0$, presented in Example 8.1. ■

Box 8.2 Detailed description for the second order correction $|n^{(2)}\rangle$ of the vector $|n\rangle$

To obtain the second order correction $|n^{(2)}\rangle$ of the vector $|n\rangle$, we apply the bra $\langle m^{(0)}|$ to Eq. (8.11), from the left. It then reads as:

$$\begin{aligned} \langle m^{(0)}| \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(2)}\rangle &= -\langle m^{(0)}|\hat{\mathcal{W}}|n^{(1)}\rangle + E_n^{(1)}\langle m^{(0)}|n^{(1)}\rangle \\ &\quad + E_n^{(2)}\langle m^{(0)}|n^{(0)}\rangle. \end{aligned} \quad (8.60)$$

The vectors of $\hat{\mathcal{H}}_0$ are orthogonal, and therefore, $\langle m^{(0)}|n^{(0)}\rangle = 0$. Furthermore, we know $E_n^{(1)}$ (Eq. (8.33)) and $\langle m^{(0)}|n^{(1)}\rangle$ (Eq. (8.39)). Thus, the above equation can be rewritten as:

$$\langle m^{(0)}|n^{(2)}\rangle = \frac{\langle m^{(0)}|\hat{\mathcal{W}}|n^{(1)}\rangle}{E_n^{(0)} - E_m^{(0)}} - \frac{\langle n^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle}{(E_n^{(0)} - E_m^{(0)})^2} \frac{\langle m^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle}{(E_n^{(0)} - E_m^{(0)})}. \quad (8.61)$$

Analogous to what was done in Box 8.1, we apply $|m^{(0)}\rangle$ to the above equation from the left and sum over all states m , except n :

$$\begin{aligned} \sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)}|n^{(2)}\rangle \\ = \sum_{m \neq n} \left[\frac{\langle m^{(0)}|\hat{\mathcal{W}}|n^{(1)}\rangle}{E_n^{(0)} - E_m^{(0)}} - \frac{\langle n^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle}{(E_n^{(0)} - E_m^{(0)})^2} \langle m^{(0)}|\hat{\mathcal{W}}|n^{(0)}\rangle \right] |m^{(0)}\rangle. \end{aligned} \quad (8.62)$$

Let us now focus our attention on the term on the left-hand side of the above equation:

$$\begin{aligned}
 & \sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(2)}\rangle \\
 &= \sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(2)}\rangle + |n^{(0)}\rangle \langle n^{(0)} | n^{(2)}\rangle - |n^{(0)}\rangle \langle n^{(0)} | n^{(2)}\rangle \\
 &= \left(\sum_m |m^{(0)}\rangle \langle m^{(0)} | \right) |n^{(2)}\rangle - |n^{(0)}\rangle \langle n^{(0)} | n^{(2)}\rangle \\
 &= |n^{(2)}\rangle - |n^{(0)}\rangle \langle n^{(0)} | n^{(2)}\rangle. \tag{8.63}
 \end{aligned}$$

After inspecting Eq. (8.16), we determine that:

$$\langle n^{(2)} | n^{(0)}\rangle = \langle n^{(0)} | n^{(2)}\rangle = -\frac{1}{2} \langle n^{(1)} | n^{(1)}\rangle, \tag{8.64}$$

and therefore:

$$\sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(2)}\rangle = |n^{(2)}\rangle + \frac{1}{2} |n^{(0)}\rangle \langle n^{(1)} | n^{(1)}\rangle. \tag{8.65}$$

We now need to determine the term $\langle n^{(1)} | n^{(1)}\rangle$, and for that, we must recall Eq. (8.34):

$$\begin{aligned}
 \langle n^{(1)} | n^{(1)}\rangle &= \sum_{m \neq n} \frac{\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)}\rangle^*}{E_n^{(0)} - E_m^{(0)}} \frac{\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} \langle m^{(0)} | m^{(0)}\rangle \\
 &= \sum_{m \neq n} \frac{|\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)}\rangle|^2}{\left(E_n^{(0)} - E_m^{(0)}\right)^2}
 \end{aligned} \tag{8.66}$$

where $\langle m^{(0)} | m^{(0)}\rangle = 1$. Substituting the above result into Eq. (8.65), we find:

$$\sum_{m \neq n} |m^{(0)}\rangle \langle m^{(0)} | n^{(2)}\rangle = |n^{(2)}\rangle + \frac{1}{2} \sum_{m \neq n} \frac{|\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)}\rangle|^2}{\left(E_n^{(0)} - E_m^{(0)}\right)^2} |n^{(0)}\rangle. \tag{8.67}$$

Let us now pause and understand what we are doing: we were developing Eq. (8.62) when we suspended the process to discuss the left-hand side of the mentioned equation. After a few steps, we obtained Eq. (8.67). Now, we will substitute Eq. (8.67) into Eq. (8.62) and continue

with our goal: to obtain the second order correction $|n^{(2)}\rangle$ of the state $|n\rangle$. Thus, we have:

$$|n^{(2)}\rangle = -\frac{1}{2} \sum_{m \neq n} \frac{|\langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle|^2}{(E_n^{(0)} - E_m^{(0)})^2} |n^{(0)}\rangle \quad (8.68)$$

$$+ \sum_{m \neq n} \left[\frac{\langle m^{(0)} | \hat{\mathcal{W}} | n^{(1)} \rangle}{E_n^{(0)} - E_m^{(0)}} - \frac{\langle n^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle}{(E_n^{(0)} - E_m^{(0)})^2} \langle m^{(0)} | \hat{\mathcal{W}} | n^{(0)} \rangle \right] |m^{(0)}\rangle.$$

Note, however, that the equation above is not yet fully written in terms of the basis vectors of the unperturbed Hamiltonian $\hat{\mathcal{H}}_0$. To address this point, we must incorporate the previously obtained result for $|n^{(1)}\rangle$ (Eq. (8.34)). Thus, we obtain the final result:

$$|n^{(2)}\rangle = -\frac{1}{2} \sum_{m \neq n} \frac{|\mathcal{W}_{mn}|^2}{(E_{nm})^2} |n^{(0)}\rangle - \sum_{m \neq n} \frac{\mathcal{W}_{nn} \mathcal{W}_{mn}}{(E_{nm})^2} |m^{(0)}\rangle$$

$$+ \sum_{m \neq n} \sum_{l \neq n} \frac{\mathcal{W}_{ml} \mathcal{W}_{ln}}{E_{nm} E_{nl}} |m^{(0)}\rangle \quad (8.69)$$

where $\mathcal{W}_{ab} = \langle a^{(0)} | \hat{\mathcal{W}} | b^{(0)} \rangle$ and $E_{ab} = E_a^{(0)} - E_b^{(0)}$.

8.2 Degenerate case

In this case, there is more than one unperturbed state for the same energy eigenvalue $E_n^{(0)}$. This condition immediately creates a problem, as there is a divergence in the denominator of Eq. (8.48). Thus, a slight adjustment is required in how we obtain the energy corrections. Therefore, let us consider that the unperturbed and degenerate Hamiltonian is written as:

$$\hat{\mathcal{H}}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle, \quad (8.70)$$

where there exists a set of vectors $|l^{(0)}\rangle$ with the same energy eigenvalue. We can define the projection operators as:

$$\hat{\mathbb{A}} = \sum_{k \notin B} |k^{(0)}\rangle \langle k^{(0)}| \quad \text{and} \quad \hat{\mathbb{B}} = \sum_{l \in B} |l^{(0)}\rangle \langle l^{(0)}|, \quad (8.71)$$

where B is the domain of the degeneracy. Naturally:

$$\hat{\mathbb{A}} + \hat{\mathbb{B}} = \mathbb{1} = \sum_n |n^{(0)}\rangle \langle n^{(0)}|. \quad (8.72)$$

■ Example 8.5

Let us consider the degenerate Hamiltonian below and (i) write its basis vectors $|n^{(0)}\rangle$ and (ii) the projection operators, taking into account the respective degeneracy. Thus:

$$\hat{H}_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (8.73)$$

where the basis vectors $|n^{(0)}\rangle$ is given by:

$$|1^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |2^{(0)}\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad |3^{(0)}\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; \quad (8.74)$$

while the corresponding zero order energies are given by:

$$E_1^{(0)} = 0, \quad E_2^{(0)} = 1, \quad E_3^{(0)} = 1. \quad (8.75)$$

Therefore, we identify that the states $|2^{(0)}\rangle$ and $|3^{(0)}\rangle$ are degenerate with energy $E_2^{(0)} = E_3^{(0)} = 1$. For simplicity, since our focus is on the calculation methodology, we are considering the Hamiltonian to be dimensionless. For this example, the two projection operators are given by:

$$\begin{aligned} \hat{A} &= |1^{(0)}\rangle\langle 1^{(0)}| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \\ \hat{B} &= |2^{(0)}\rangle\langle 2^{(0)}| + |3^{(0)}\rangle\langle 3^{(0)}| = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (8.76)$$

■

We can discuss some properties of these projection operators, namely:

Property 8.1 The square of the projection operator is the projection operator itself.

$$\hat{A}^2 = \hat{A} \quad (8.77)$$

Proof.

$$\hat{A}^2 = \sum_k |k^{(0)}\rangle \langle k^{(0)}| k^{(0)}\rangle \langle k^{(0)}| = \sum_k |k^{(0)}\rangle \langle k^{(0)}| = \hat{A}. \quad (8.78)$$

■

Property 8.2 The multiplication of complementary projection operators is zero.

$$\hat{A}\hat{B} = 0 \quad (8.79)$$

Proof.

$$\hat{A}\hat{B} = \hat{A}(\mathbb{1} - \hat{A}) = (\hat{A} - \hat{A}^2) = (\hat{A} - \hat{A}) = 0. \quad (8.80)$$

■

Property 8.3 The unperturbed Hamiltonian commutes with the projection operator.

$$[\hat{A}, \hat{\mathcal{H}}_0] = 0 \quad (8.81)$$

Proof.

$$[\hat{A}, \hat{\mathcal{H}}_0] = \sum_k |k^{(0)}\rangle \langle k^{(0)}| \hat{\mathcal{H}}_0 - \sum_k \hat{\mathcal{H}}_0 |k^{(0)}\rangle \langle k^{(0)}| = E_k^{(0)} (\hat{A} - \hat{A}) = 0. \quad (8.82)$$

■

Analogous to what we did for the non-degenerate case, the total Hamiltonian of the system can be written as:

$$\hat{\mathcal{H}}|n\rangle = E_n|n\rangle, \quad \text{where} \quad \hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \lambda\hat{\mathcal{V}}. \quad (8.83)$$

We need to ensure that in the limit $\lambda \rightarrow 0$, Eq. (8.83) recovers Eq. (8.70), thus guaranteeing that the vectors $|n\rangle$ and energies E_n continuously approach the known unperturbed solution. For this purpose, we consider the expansions of the vectors and energies similarly to what we did in the non-degenerate case (Eqs. (8.6) and (8.7)).

Objective The objective of this section is, therefore, to approximately obtain the eigenvalues and vectors of the problem – taking into account the degeneracy present in the unperturbed Hamiltonian.

8.2.1 First order correction to energy

Let us consider Eq. (8.83) and the projection operators discussed earlier:

$$(\hat{\mathcal{H}}_0 + \lambda \hat{\mathcal{W}} - E_n)(\hat{\mathbb{A}} + \hat{\mathbb{B}})|n\rangle = 0. \quad (8.84)$$

At this point, it is appropriate to define new bases for the problem:

$$\hat{\mathbb{A}}|n\rangle = |na\rangle \quad \text{and} \quad \hat{\mathbb{B}}|n\rangle = |nb\rangle, \quad (8.85)$$

where a and b indicate the domain of degeneracy. For these new bases, we have the following properties:

Property 8.4

$$\hat{\mathbb{B}}|nb\rangle = |nb\rangle \quad (8.86)$$

Proof.

$$\hat{\mathbb{B}}|nb\rangle = \hat{\mathbb{B}}^2|n\rangle = \hat{\mathbb{B}}|n\rangle = |nb\rangle. \quad (8.87)$$

■

Property 8.5

$$\hat{\mathbb{A}}|nb\rangle = 0 \quad (8.88)$$

Proof.

$$\hat{\mathbb{A}}|nb\rangle = \hat{\mathbb{A}}\hat{\mathbb{B}}|n\rangle = 0. \quad (8.89)$$

■

After introducing these new bases, we can return to Eq. (8.84) and rewrite it as:

$$\hat{\mathcal{H}}_0[|na\rangle + |nb\rangle] + \lambda \hat{\mathcal{W}}[|na\rangle + |nb\rangle] - E_n[|na\rangle + |nb\rangle] = 0. \quad (8.90)$$

Let us apply the projection operator $\hat{\mathbb{A}}$, from the left, to Eq. (8.90). We thus find:

$$\left[(E_n - \hat{\mathcal{H}}_0)\hat{\mathbb{A}} - \lambda \hat{\mathcal{W}}_A \right] |na\rangle = \lambda \hat{\mathbb{A}} \hat{\mathcal{W}} \hat{\mathbb{B}} |nb\rangle, \quad (8.91)$$

where $\hat{\mathcal{W}}_A = \hat{\mathbb{A}}\hat{\mathcal{W}}\hat{\mathbb{A}}$ is the A -reduced perturbation. In Example 8.6, we will examine the effect of the projection operator on the operation $\hat{\mathbb{A}}\hat{\mathcal{W}}\hat{\mathbb{A}}$. We can perform the same operation as before, but now applying the projection operator $\hat{\mathbb{B}}$, from the left, to Eq. (8.90). Thus, we have:

$$\left[(E_n - \hat{\mathcal{H}}_0)\hat{\mathbb{B}} - \lambda\hat{\mathcal{W}}_B \right] |nb\rangle = \lambda\hat{\mathbb{B}}\hat{\mathcal{W}}\hat{\mathbb{A}}|na\rangle. \quad (8.92)$$

Substituting Eq. (8.91) into Eq. (8.92), we obtain a system for $|nb\rangle$:

$$\left[(E_{nb} - \hat{\mathcal{H}}_0)\hat{\mathbb{B}} - \lambda\hat{\mathcal{W}}_B \right] |nb\rangle = \lambda^2\hat{\mathbb{B}}\hat{\mathcal{W}}\hat{\mathbb{A}} \left[(E_{nb} - \hat{\mathcal{H}}_0)\hat{\mathbb{A}} - \lambda\hat{\mathcal{W}}_A \right]^{-1} \hat{\mathbb{A}}\hat{\mathcal{W}}\hat{\mathbb{B}}|nb\rangle. \quad (8.93)$$

We place the index ‘b’ on the energy (E_{nb}) to indicate that it refers to the degeneracy domain B . Let us now consider the expansion of the energy and vectors, similar to Eqs. (8.6) and (8.7). Thus:

$$|nb\rangle = |nb^{(0)}\rangle + \lambda|nb^{(1)}\rangle + \dots \quad \text{and} \quad E_{nb} = E_{nb}^{(0)} + \lambda E_{nb}^{(1)} + \lambda^2 E_{nb}^{(2)} + \dots \quad (8.94)$$

Substituting these expansions into Eq. (8.93) and considering only terms up to λ^1 , we find:

$$\left[(E_{nb}^{(0)} + \lambda E_{nb}^{(1)} - \hat{\mathcal{H}}_0)\hat{\mathbb{B}} - \lambda\hat{\mathcal{W}}_B \right] \left[|nb^{(0)}\rangle + \lambda|nb^{(1)}\rangle \right] = 0. \quad (8.95)$$

From the equation above, we can collect the terms of the same order in λ :

$$\lambda^0 : \quad \left(\hat{\mathcal{H}}_0 - E_{nb}^{(0)} \right) \hat{\mathbb{B}} |nb^{(0)}\rangle = 0 \quad (8.96)$$

$$\lambda^1 : \quad \left(\hat{\mathcal{W}}_B - E_{nb}^{(1)} \right) \hat{\mathbb{B}} |nb^{(0)}\rangle = 0 \quad (8.97)$$

Eq. (8.96) shows that the eigenvalues of the unperturbed Hamiltonian in the basis vectors $\{|nb^{(0)}\rangle\}$ are the energy eigenvalues $E_{nb}^{(0)} = E_b$ of the degenerate domain B . Through Eq. (8.97), and using the same basis vectors $\{|nb^{(0)}\rangle\}$, it is possible to determine the first energy correction $E_{nb}^{(1)}$, which are the eigenvalues of the B -reduced perturbation $\hat{\mathcal{W}}_B$. It is important to emphasize that we chose the domain B as degenerate, but we could have chosen the domain A . The equations obtained thus far refer to domain B , but they can (and should) be applied to all domains of the problem.

■ Example 8.6

Let us consider the information from Example 8.5 and, in addition,

the perturbation:

$$\hat{\mathcal{W}} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}. \quad (8.98)$$

From Eq. (8.97), it is possible to calculate the vectors $|nb^{(0)}\rangle$ for domain B , as well as the vectors $|na^{(0)}\rangle$ for domain A . We will also calculate the respective first order energies $E_{nb}^{(1)}$ (for domain B) and $E_{na}^{(1)}$ (for domain A).

First, let us write the B -reduced perturbation matrix $\hat{\mathcal{W}}_B$:

$$\hat{\mathcal{W}}_B = \hat{\mathbb{B}}\hat{\mathcal{W}}\hat{\mathbb{B}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}. \quad (8.99)$$

Note that this procedure has the effect of ‘isolating’ the respective domain of the perturbation matrix $\hat{\mathcal{W}}$. Using $\hat{\mathcal{W}}_B$ in Eq. (8.97), we find the corresponding eigenvalues:

$$E_{1b}^{(1)} = -2 \quad \text{and} \quad E_{2b}^{(1)} = +2, \quad (8.100)$$

as well as the vectors, given by:

$$\begin{aligned} |1b^{(0)}\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[-|2^{(0)}\rangle + |3^{(0)}\rangle \right] \\ |2b^{(0)}\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[|2^{(0)}\rangle + |3^{(0)}\rangle \right]. \end{aligned} \quad (8.101)$$

The same procedure can be applied to domain A , where Equations 8.96 and 8.97 are valid. Thus:

$$\hat{\mathcal{W}}_A = \hat{\mathbb{A}}\hat{\mathcal{W}}\hat{\mathbb{A}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (8.102)$$

and, using $\hat{\mathcal{W}}_A$ in Eq. (8.97), we find a single eigenvalue for this case:

$$E_{1a}^{(1)} = 1, \quad (8.103)$$

as well as the corresponding vector:

$$|1a^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = |1^{(0)}\rangle. \quad (8.104)$$

Naturally, due to Eq. (8.96), the zero order energy is the same as that discussed in Example 8.5. Using Eq. (8.96), we therefore obtain:

$$E_{1a}^{(0)} = 0 \quad E_{1b}^{(0)} = 1 \quad E_{2b}^{(0)} = 1. \quad (8.105)$$

■

8.2.2 First order correction to vectors

Just as in Eq. (8.85), we can use the projection operators to write:

$$|n\rangle = (\hat{A} + \hat{B})|n\rangle = |na\rangle + |nb\rangle. \quad (8.106)$$

However, we know that:

$$|n\rangle \approx [|na^{(0)}\rangle + |nb^{(0)}\rangle] + \lambda [|na^{(1)}\rangle + |nb^{(1)}\rangle] + \dots \quad (8.107)$$

Comparing Eqs. (8.106) and (8.107), results in:

$$|nb\rangle \approx |nb^{(0)}\rangle + \lambda [|na^{(1)}\rangle + |nb^{(1)}\rangle] + \dots \quad (8.108)$$

It is important to note that the above equation ensures the necessary limit in which $\lambda \rightarrow 0$ recovers the non-perturbed case, that is:

$$|nb\rangle \xrightarrow{\lambda \rightarrow 0} |nb^{(0)}\rangle. \quad (8.109)$$

Note also that $|nb\rangle$ carries a mixture of states. This fact will become clearer in Example 8.7. In Box 8.3, we detail how to determine $|na^{(1)}\rangle$ and $|nb^{(1)}\rangle$ in order to write the vector $|nb\rangle$ up to first order using Eq. (8.108). Thus:

$$|nb\rangle \approx |nb^{(0)}\rangle + \lambda \sum_{k \notin B} \left\{ \frac{\langle k^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle}{E_b - E_k^{(0)}} |k^{(0)}\rangle + \sum_{m \neq n} \frac{\langle mb^{(0)} | \hat{\mathcal{W}} | k^{(0)} \rangle}{(E_{nb}^{(1)} - E_{mb}^{(1)})} \frac{\langle k^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle}{(E_b - E_k^{(0)})} |mb^{(0)}\rangle \right\} \quad (8.110)$$

It is important to emphasize that we chose domain B as degenerate, but we could have chosen domain A . The equations obtained thus far refer to domain B , but they can (and should) be applied to all domains of the problem.

Box 8.3 Detailed description of how to determine the vectors $|na^{(1)}\rangle$ and $|nb^{(1)}\rangle$

Let us determine $|nb^{(1)}\rangle$. For this purpose, we will return to Eq. (8.93) and rewrite it. Thus:

$$\hat{\mathcal{H}}|nb\rangle = E_{nb}|nb\rangle \quad (8.111)$$

where

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \lambda \hat{\mathcal{W}}, \quad \text{and} \quad \hat{\mathcal{H}}_0 = \hat{\mathcal{H}}_0 + \lambda \hat{\mathcal{W}}_B, \quad (8.112)$$

and

$$\hat{\mathcal{W}} = \lambda \hat{\mathbb{B}} \hat{\mathcal{W}} \hat{\mathbb{A}} \left[(E_{nb} - \hat{\mathcal{H}}_0) \hat{\mathbb{A}} - \lambda \hat{\mathcal{W}}_A \right]^{-1} \hat{\mathbb{A}} \hat{\mathcal{W}} \hat{\mathbb{B}} \approx \lambda \frac{\hat{\mathbb{B}} \hat{\mathcal{W}} \hat{\mathbb{A}} \hat{\mathcal{W}} \hat{\mathbb{B}}}{E_b - E_a}. \quad (8.113)$$

Note that Eq. (8.93) does not involve any approximation, and therefore, neither does Eq. (8.111) – it is merely a rearrangement of terms in order to obtain an eigenvalue equation for the basis vectors $\{|nb\rangle\}$. After writing Eq. (8.111), it is possible to verify that the new term $\hat{\mathcal{H}}_0$ is diagonal for the basis vectors $\{|nb^{(0)}\rangle\}$. In addition, this term is non-degenerate due to the term $\hat{\mathcal{W}}_B$. Thus, we can apply the non-degenerate formalism (Section 8.1) to find the first order correction to the vector $|nb\rangle$, as follows:

$$|nb^{(1)}\rangle = \sum_{m \neq n} \frac{\langle mb^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle}{(E_b + \lambda E_{nb}^{(1)}) - (E_b + \lambda E_{mb}^{(1)})} |mb^{(0)}\rangle. \quad (8.114)$$

To move forward, the numerator of the above equation needs to be explored:

$$\begin{aligned} \langle mb^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle &\approx \lambda \langle mb^{(0)} | \frac{\hat{\mathcal{W}} \hat{\mathbb{A}} \hat{\mathcal{W}}}{E_b - E_a} | nb^{(0)} \rangle \\ &\approx \lambda \sum_{k \notin B} \langle mb^{(0)} | \hat{\mathcal{W}} | k^{(0)} \rangle \frac{\langle k^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle}{E_b - E_k^{(0)}}. \end{aligned} \quad (8.115)$$

Substituting Eq. (8.115) into Eq. (8.114), we find:

$$|nb^{(1)}\rangle = \sum_{k \notin B} \sum_{m \neq n} \frac{\langle mb^{(0)} | \hat{\mathcal{W}} | k^{(0)} \rangle}{(E_{nb}^{(1)} - E_{mb}^{(1)})} \frac{\langle k^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle}{(E_b - E_k^{(0)})} |mb^{(0)}\rangle \quad (8.116)$$

We also need to determine $|na^{(1)}\rangle$. For this purpose, we will refer to Eq. (8.91):

$$|na\rangle = \lambda \left[(E_n - \hat{\mathcal{H}}_0) \hat{\mathbb{A}} - \lambda \hat{\mathcal{W}}_A \right]^{-1} \hat{\mathbb{A}} \hat{\mathcal{W}} \hat{\mathbb{B}} |nb\rangle. \quad (8.117)$$

Expanding vectors and energy, we obtain:

$$\left[|na^{(0)}\rangle + \lambda |na^{(1)}\rangle \right] = \lambda \frac{\hat{\mathbb{A}} \hat{\mathcal{W}} \hat{\mathbb{B}}}{E_b - E_a} \left[|nb^{(0)}\rangle + \lambda |nb^{(1)}\rangle \right], \quad (8.118)$$

where we can collect only the terms in λ^1 :

$$\lambda^1: \quad |na^{(1)}\rangle = \frac{\hat{\mathbb{A}} \hat{\mathcal{W}}}{E_b - E_a} |nb^{(0)}\rangle. \quad (8.119)$$

The equation above can then be rewritten as:

$$|na^{(1)}\rangle = \sum_{k \notin B} \frac{\langle k^{(0)} | \hat{\mathcal{W}} | nb^{(0)} \rangle}{E_b - E_k^{(0)}} |k^{(0)}\rangle \quad (8.120)$$

■ Example 8.7

Let us continue with Examples 8.5 and 8.6, where we will now calculate the vectors up to first order, in accordance with Eq. (8.110):

$$|1a\rangle \approx |1a^{(0)}\rangle + \lambda \left\{ \frac{\langle 2^{(0)} | \hat{\mathcal{W}} | 1a^{(0)} \rangle}{(-1)} |2^{(0)}\rangle + \frac{\langle 3^{(0)} | \hat{\mathcal{W}} | 1a^{(0)} \rangle}{(-1)} |3^{(0)}\rangle \right\} \quad (8.121)$$

$$\approx |1^{(0)}\rangle - \lambda |2^{(0)}\rangle - 2\lambda |3^{(0)}\rangle, \quad (8.122)$$

where $|1a^{(0)}\rangle = |1^{(0)}\rangle$, in accordance with Eq. (8.104). Similarly to the calculation of the vector $|1a\rangle$, we can obtain:

$$|1b\rangle \approx |1b^{(0)}\rangle + \lambda \left\{ \frac{\langle 1^{(0)} | \hat{\mathcal{W}} | 1b^{(0)} \rangle}{(1)} |1^{(0)}\rangle + \frac{\langle 2b^{(0)} | \hat{\mathcal{W}} | 1^{(0)} \rangle}{(-2-2)} \frac{\langle 1^{(0)} | \hat{\mathcal{W}} | 1b^{(0)} \rangle}{(1)} |2b^{(0)}\rangle \right\}. \quad (8.123)$$

Remember that $|1b^{(0)}\rangle$ and $|2b^{(0)}\rangle$ were previously determined (Eq. (8.101)). In this way, we can rewrite the vector $|1b\rangle$ in terms

of the basis vectors $|n^{(0)}\rangle$:

$$|1b\rangle \approx \frac{\sqrt{2}}{2}\lambda|1^{(0)}\rangle - \frac{\sqrt{2}}{16}(8+3\lambda)|2^{(0)}\rangle + \frac{\sqrt{2}}{16}(8-3\lambda)|3^{(0)}\rangle. \quad (8.124)$$

Finally, we can also determine the vector $|2b\rangle$, given by:

$$\begin{aligned} |2b\rangle &\approx |2b^{(0)}\rangle \\ &+ \lambda \left\{ \frac{\langle 1^{(0)}|\hat{\mathcal{W}}|2b^{(0)}\rangle}{(1)}|1^{(0)}\rangle + \frac{\langle 1b^{(0)}|\hat{\mathcal{W}}|1^{(0)}\rangle}{[2-(-2)]} \frac{\langle 1^{(0)}|\hat{\mathcal{W}}|2b^{(0)}\rangle}{(1)}|1b^{(0)}\rangle \right\} \\ &\approx \frac{3\sqrt{2}}{2}\lambda|1^{(0)}\rangle + \frac{\sqrt{2}}{16}(8-3\lambda)|2^{(0)}\rangle + \frac{\sqrt{2}}{16}(8+3\lambda)|3^{(0)}\rangle. \end{aligned} \quad (8.125)$$

■

8.2.3 Second order correction to the energy

To obtain the second order correction to the energy, we must recall Eq. (8.92), considering the expansion of the energy (up to second order) and vectors (up to first order):

$$\left\{ \left(E_{nb}^{(1)} + \lambda E_{nb}^{(2)} \right) \hat{\mathbb{B}} - \hat{\mathcal{W}}_B \right\} \left[|nb^{(0)}\rangle + \lambda |nb^{(1)}\rangle \right] = \hat{\mathbb{B}} \hat{\mathcal{W}} \hat{\mathbb{A}} \left[|na^{(0)}\rangle + \lambda |na^{(1)}\rangle \right]. \quad (8.126)$$

Collecting only the terms in λ^1 results in:

$$\lambda^1 : \quad E_{nb}^{(2)}|nb^{(0)}\rangle + E_{nb}^{(1)}|nb^{(1)}\rangle = \hat{\mathbb{B}} \left[\hat{\mathcal{W}}|na^{(1)}\rangle + \hat{\mathcal{W}}|nb^{(1)}\rangle \right]. \quad (8.127)$$

Let us apply $\langle nb^{(0)}|$, from the left, to the above equation. We then have:

$$E_{nb}^{(2)} = \langle nb^{(0)}|\hat{\mathcal{W}}|na^{(1)}\rangle + \langle nb^{(0)}|\hat{\mathcal{W}}|nb^{(1)}\rangle, \quad (8.128)$$

since $\langle nb^{(0)}|nb^{(1)}\rangle = 0$ (see Box 8.3 for further details). Eq. (8.128) has two terms. Let us analyze the second term: $\langle nb^{(0)}|\hat{\mathcal{W}}|nb^{(1)}\rangle = \langle nb^{(0)}|\hat{\mathcal{W}}_B|nb^{(1)}\rangle$, where $\hat{\mathcal{W}}_B$ is diagonal in the basis vectors $\{|nb^{(0)}\rangle\}$. We know from Eq. (8.116) that $|nb^{(1)}\rangle$ is a linear combination of the vectors $|mb^{(0)}\rangle$, except for $|nb^{(0)}\rangle$. Therefore, the second term of Eq. (8.128) excludes the main diagonal terms of $\hat{\mathcal{W}}_B$ – precisely the non-zero elements. Thus, the mentioned term is zero. Therefore, we have:

$$E_{nb}^{(2)} = \langle nb^{(0)}|\hat{\mathcal{W}}|na^{(1)}\rangle \quad (8.129)$$

and, by retrieving Eq. (8.120), we obtain:

$$E_{nb}^{(2)} = \sum_{k \notin B} \frac{|\langle k^{(0)} | \hat{W} | nb^{(0)} \rangle|^2}{E_b - E_k^{(0)}} \quad (8.130)$$

Thus, the second order energy correction is a sum over the states outside of the considered degeneracy domain.

■ Example 8.8

Analogous to the previous examples, let us revisit Examples 8.5–8.7 and calculate the second order energy correction for the problem we are working on:

$$E_{1a}^{(2)} = \frac{|\langle 2^{(0)} | \hat{W} | 1a^{(0)} \rangle|^2}{(-1)} + \frac{|\langle 3^{(0)} | \hat{W} | 1a^{(0)} \rangle|^2}{(-1)} = -5, \quad (8.131)$$

$$E_{1b}^{(2)} = \frac{|\langle 1^{(0)} | \hat{W} | 1b^{(0)} \rangle|^2}{(1)} = \frac{1}{2}, \quad (8.132)$$

$$E_{2b}^{(2)} = \frac{|\langle 1^{(0)} | \hat{W} | 2b^{(0)} \rangle|^2}{(1)} = \frac{9}{2}. \quad (8.133)$$

From the results above, it is possible then to write the final energy of the system studied in these examples, in powers of λ , given by:

$$E_{1a} = \lambda - 5\lambda^2, \quad E_{1b} = 1 - 2\lambda + \frac{\lambda^2}{2} \quad \text{and} \quad E_{2b} = 1 + 2\lambda + \frac{9\lambda^2}{2}. \quad (8.134)$$

■

Application 8.A van Vleck susceptibility

The results presented from this point onward were, of course, developed by van Vleck, serving as an important tool for cases where it is important to know the magnetic susceptibility of the material, however, it is not possible to exactly determine the energy spectrum of the system. The idea is to use Perturbation Theory to obtain the first and/or second energy corrections and thus write the corresponding magnetic susceptibility.

We begin by considering the Zeeman energy of the i -state – more precisely, the interaction energy of a magnetic moment $\vec{\mu}_i$ with the magnetic

field $\vec{B}$, given by $E_i = -\vec{\mu}_i \cdot \vec{B}$, which can be rewritten as $E_i = -\sum_v (\mu_i)_v B_v$, where $v = (x, y, z)$. Note the magnetic moment can be written as:

$$(\mu_i)_v = -\frac{\partial E_i}{\partial B_v}. \quad (8.A.1)$$

To simplify the problem, let us consider $\vec{B} = B_u \hat{u}$, that is, $\vec{B}$ pointing along a specific axis $\hat{u}$ of the crystal. This assumption results in $E_i = -(\mu_i)_u B_u$.

From the course of statistical mechanics, the thermal average $\langle \hat{\mathcal{O}} \rangle$ of a physical quantity must be written as:

$$\langle \hat{\mathcal{O}} \rangle = \frac{\sum_i \mathcal{O}_i e^{-E_i \beta}}{\sum_i e^{-E_i \beta}}, \quad (8.A.2)$$

where $\beta = 1/k_B T$ and $\mathcal{O}_i$ are the eigenvalues of the operator $\hat{\mathcal{O}}$ representing the physical quantity of interest. Thus, the magnetization along a given direction u should be written as:

$$\frac{M_u}{N} = \langle \hat{\mu}_u \rangle = \frac{\sum_i (\mu_i)_u e^{-E_i \beta}}{\sum_i e^{-E_i \beta}}. \quad (8.A.3)$$

Van Vleck's idea was to expand the energy spectrum in powers of the magnetic field B_u :

$$E_i = E_i^{(0)} + E_i^{(1)} B_u + E_i^{(2)} B_u^2 + \dots \quad (8.A.4)$$

Consequently, the magnetic moment (through Eq. (8.A.1)) can be written as:

$$(\mu_i)_u = -E_i^{(1)} - 2E_i^{(2)} B_u - \dots \quad (8.A.5)$$

Another approximation considered is $B_u/k_B T \ll 1$. Thus:

$$e^{-E_i \beta} = \exp \left[- \left(E_i^{(0)} + E_i^{(1)} B_u \right) \beta \right] = e^{-E_i^{(0)} \beta} \left[1 - E_i^{(1)} B_u \beta \right]. \quad (8.A.6)$$

Substituting these approximations into Eq. (8.A.3), we find:

$$\frac{M_u}{N} = \langle \hat{\mu}_u \rangle = \frac{\sum_i \left(-E_i^{(1)} - 2E_i^{(2)} B_u \right) e^{-E_i^{(0)} \beta} \left[1 - E_i^{(1)} B_u \beta \right]}{\sum_i e^{-E_i^{(0)} \beta} \left[1 - E_i^{(1)} B_u \beta \right]}. \quad (8.A.7)$$

To simplify the result above, two points must be considered: (i) neglect terms of order B_u^2 or higher, and (ii) zero-field magnetization is zero. This latter consideration leads us to write:

$$\frac{M_u}{N} (B_u = 0) = -\frac{\sum_i E_i^{(1)} e^{-E_i^{(0)} \beta}}{\sum_i e^{-E_i^{(0)} \beta}} = 0 \quad \Rightarrow \quad \sum_i E_i^{(1)} e^{-E_i^{(0)} \beta} = 0 \quad (8.A.8)$$

Thus:

$$\frac{M_u}{N} = B_u \frac{\sum_i \left[\left(E_i^{(1)} \right)^2 \beta - 2E_i^{(2)} \right] e^{-E_i^{(0)} \beta}}{\sum_i e^{-E_i^{(0)} \beta}}. \quad (8.A.9)$$

However, we are looking for the susceptibility, given by:

$$\chi_{uv} = \lim_{B_v \rightarrow 0} \mu_0 \frac{dM_u}{dB_v}. \quad (8.A.10)$$

Using the equation above, we then obtain van Vleck's susceptibility (for the particular case $u = v$):

$$\chi_{uu} = N\mu_0 \frac{\sum_i \left[\left(E_i^{(1)} \right)^2 \beta - 2E_i^{(2)} \right] e^{-E_i^{(0)} \beta}}{\sum_i e^{-E_i^{(0)} \beta}} \quad (8.A.11)$$

For the van Vleck's susceptibility, the energy corrections ($E_i^{(1)}$ and $E_i^{(2)}$) are obtained through the Perturbation Theory studied in this chapter. Note, however, that the parameter λ used earlier in the chapter is now B_u in this application, meaning the energy expansion is done in powers of B_u . The energy corrections $E_i^{(1)}$ and $E_i^{(2)}$ are calculated according to Eqs. (8.33) and (8.48) for the non-degenerate case, and according to Eqs. (8.97) and (8.130) for the degenerate case.

★ **John van Vleck (1899–1980):** Born in Middletown, United States. He developed quantum theories for magnetism. In 1977, he received the Nobel Prize in Physics for his research on ferromagnetism and superconductivity.

■ Example 8.9

In this example, let us consider the following Hamiltonian:

$$\hat{\mathcal{H}} = D\hat{J}_z^2 + \mu_B g_u \hat{J}_u B_u, \quad (8.A.12)$$

where $u = x, y, z$. See Section 7.1 where, in those Hamiltonians, for simplicity, the angular momentum operator is dimensionless, and the parameter regulating the intensity of the term (e.g., D) has the dimension of energy. The same procedure is adopted in this example. The first term of the equation above refers to local magnetocrystalline anisotropy, while the second term is the Zeeman contribution.² We are considering an index u in the Landé factor

²See Appendix 7.A for more details regarding the Zeeman term.

to account for the system's anisotropy. We can identify these terms in the Hamiltonian as those from Perturbation Theory:

$$\hat{\mathcal{H}}_0 = D\hat{J}_z^2, \quad \hat{\mathcal{W}} = \mu_B g_u \hat{J}_u \quad \text{and} \quad \lambda = B_u. \quad (8.A.13)$$

In this example, let us consider a total angular momentum $j = 3/2$, where its operators $\hat{J}_x$, $\hat{J}_y$, and $\hat{J}_z$ can be obtained through the procedure in Section 6.4:

$$\hat{J}_x = \begin{pmatrix} 0 & 0 & \sqrt{3}/2 & 0 \\ 0 & 0 & 0 & \sqrt{3}/2 \\ \sqrt{3}/2 & 0 & 0 & 1 \\ 0 & \sqrt{3}/2 & 1 & 0 \end{pmatrix}, \quad (8.A.14)$$

$$\hat{J}_y = i \begin{pmatrix} 0 & 0 & -\sqrt{3}/2 & 0 \\ 0 & 0 & 0 & \sqrt{3}/2 \\ \sqrt{3}/2 & 0 & 0 & -1 \\ 0 & -\sqrt{3}/2 & 1 & 0 \end{pmatrix}, \quad (8.A.15)$$

$$\hat{J}_z = \begin{pmatrix} 3/2 & 0 & 0 & 0 \\ 0 & -3/2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & -1/2 \end{pmatrix}. \quad (8.A.16)$$

These matrices were constructed with the following vectors $|j, m\rangle = |m\rangle$:

$$|+3/2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |-3/2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |+1/2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |-1/2\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad (8.A.17)$$

in such a way that the unperturbed Hamiltonian can be written as:

$$\hat{\mathcal{H}}_0 = \frac{D}{4} \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (8.A.18)$$

Note that we can order the basis vectors of the unperturbed Hamiltonian in different ways, and therefore, Eqs. (8.A.17) were chosen in order to have the degenerate eigenvalues grouped together.

Regarding the perturbation $\hat{\mathcal{W}}$, it may depend on the direction in which the magnetic field is applied. In this example, let's consider

$B_u = B_x$:

$$\hat{\mathcal{W}} = \mu_B g_x \hat{J}_x = \mu_B g_x \begin{pmatrix} 0 & 0 & \sqrt{3}/2 & 0 \\ 0 & 0 & 0 & \sqrt{3}/2 \\ \sqrt{3}/2 & 0 & 0 & 1 \\ 0 & \sqrt{3}/2 & 1 & 0 \end{pmatrix}. \quad (8.A.19)$$

From Eq. (8.97) and using the A -reduced matrix:

$$\hat{\mathcal{W}}_A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (8.A.20)$$

we can obtain the first energy correction, as well as the vectors $|na^{(0)}\rangle$. The vectors are:

$$|1a^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = | + 3/2 \rangle \quad \text{and} \quad |2a^{(0)}\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = | - 3/2 \rangle, \quad (8.A.21)$$

while the first energy corrections are:

$$E_{1a}^{(1)} = 0 \quad \text{and} \quad E_{2a}^{(1)} = 0. \quad (8.A.22)$$

The same procedure can be applied to the B degeneracy domain. Using the B -reduced matrix:

$$\hat{\mathcal{W}}_B = \mu_B g_x \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (8.A.23)$$

we obtain:

$$|1b^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} [- | + 1/2 \rangle + | - 1/2 \rangle] \quad (8.A.24)$$

and

$$|2b^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} [| + 1/2 \rangle + | - 1/2 \rangle]; \quad (8.A.25)$$

while the first energy corrections are:

$$E_{1b}^{(1)} = -\mu_B g_x \quad \text{and} \quad E_{2b}^{(1)} = \mu_B g_x. \quad (8.A.26)$$

Once we know the vectors $|na^{(0)}\rangle$ and $|nb^{(0)}\rangle$, we can use Eq. (8.96) to determine the zero order energy³:

$$E_{1a}^{(0)} = E_{2a}^{(0)} = \frac{9D}{4} \quad \text{and} \quad E_{1b}^{(0)} = E_{2b}^{(0)} = \frac{D}{4}. \quad (8.A.27)$$

Finally, we need to determine the second order energy correction, and, for this purpose, we must refer to Eq. (8.130). Thus:

$$E_{1a}^{(2)} = \frac{|\langle +1/2 | \hat{\mathcal{W}} | 1a^{(0)} \rangle|^2}{(\frac{9D}{4} - \frac{D}{4})} + \frac{|\langle -1/2 | \hat{\mathcal{W}} | 1a^{(0)} \rangle|^2}{(\frac{9D}{4} - \frac{D}{4})}. \quad (8.A.28)$$

We have determined $|1a^{(0)}\rangle$ in Eq. (8.A.21), and then we conclude that:

$$E_{1a}^{(2)} = \frac{3}{8D} g_x^2 \mu_B^2. \quad (8.A.29)$$

Similarly, it is possible to write that:

$$E_{2a}^{(2)} = \frac{|\langle +1/2 | \hat{\mathcal{W}} | 2a^{(0)} \rangle|^2}{2D} + \frac{|\langle -1/2 | \hat{\mathcal{W}} | 2a^{(0)} \rangle|^2}{2D} = \frac{3}{8D} g_x^2 \mu_B^2, \quad (8.A.30)$$

where we also consider $|2a^{(0)}\rangle$ from Eq. (8.A.21). For the B domain, we find:

$$E_{1b}^{(2)} = \frac{|\langle +3/2 | \hat{\mathcal{W}} | 1b^{(0)} \rangle|^2}{-2D} + \frac{|\langle -3/2 | \hat{\mathcal{W}} | 1b^{(0)} \rangle|^2}{-2D}. \quad (8.A.31)$$

Considering $|1b^{(0)}\rangle$ obtained in Eq. (8.A.24), it is possible to rewrite the equation above as:

$$E_{1b}^{(2)} = -\frac{3}{8D} g_x^2 \mu_B^2. \quad (8.A.32)$$

Finally, we find:

$$E_{2b}^{(2)} = \frac{|\langle +3/2 | \hat{\mathcal{W}} | 2b^{(0)} \rangle|^2}{-2D} + \frac{|\langle -3/2 | \hat{\mathcal{W}} | 2b^{(0)} \rangle|^2}{-2D} = -\frac{3}{8D} g_x^2 \mu_B^2, \quad (8.A.33)$$

³This zero order energy could have been obtained previously for the $\{|j, m\rangle\}$ basis vector.

where we consider $|2b^{(0)}\rangle$ determined in Eq. (8.A.25).

The energies obtained are summarized in Table 8.1. From these values, it is then possible to use Eq. (8.A.11) to determine the van Vleck susceptibility:

$$\frac{\chi_{xx}}{N\mu_0} = \frac{(\mu_B g_x)^2}{4k_B T D} \frac{(-3k_B T e^{-2D/k_B T} + 4D + 3k_B T)}{(e^{-2D/k_B T} + 1)}. \quad (8.A.34)$$

■

TABLE 8.1 Energy spectrum corrections of the Hamiltonian (8.A.12), for the case $u = x$.

Vector	$E_{na/nb}^{(0)}$	$E_{na/nb}^{(1)}$	$E_{na/nb}^{(2)}$
$ 1a\rangle$	$9D/4$	0	$3(\mu_B g_x)^2/8D$
$ 2a\rangle$	$9D/4$	0	$3(\mu_B g_x)^2/8D$
$ 1b\rangle$	$D/4$	$-\mu_B g_x$	$-3(\mu_B g_x)^2/8D$
$ 2b\rangle$	$D/4$	$\mu_B g_x$	$-3(\mu_B g_x)^2/8D$

Application 8.B Stark effect

The Stark effect addresses the influence of a constant electric field on a Hydrogen atom. The unperturbed system was thoroughly discussed in Section 5.3, for which we recommend reviewing the mentioned section. However, a brief summary of the unperturbed Hydrogen atom is provided below:

Hamiltonian It consists of a kinetic term and a Coulomb potential:

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}. \quad (8.B.1)$$

Wave function Composed of a radial $R_{nl}(r)$ and an angular⁴ $Y_l^m(\theta, \phi)$ contributions:

$$\Psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_l^m(\theta, \phi). \quad (8.B.2)$$

The total wave function, which is the solution to the Schrödinger equation, can be found in Eq. (5.107). The quantum numbers involved are: $n = 1, 2, 3, \dots$, known as the principal quantum number; $l = 0, 1, 2, \dots, n-1$, the orbital quantum number; and finally, $-l \leq m \leq +l$, the magnetic quantum number.

⁴Spherical harmonics.

Energy The energy spectrum of the system, after solving the Schrödinger equation, is given by:

$$E_n^{(0)} = -\frac{E_0}{n^2}, \quad (8.B.3)$$

where $E_0 = 13.6$ eV is the ionization energy of the atom. This system is degenerate, as discussed in Section 5.3, because there are n^2 states $\Psi_{nlm}(r, \theta, \phi)$ with the same energy value $E_n^{(0)}$.

We can write the eigenvalue equation for the unperturbed system as:

$$\hat{\mathcal{H}}_0 |nlm\rangle = E_n^{(0)} |nlm\rangle; \quad (8.B.4)$$

while the basis vectors $\{|nlm\rangle\}$, for $n \leq 2$, can be constructed as follows:

$$\begin{aligned} |100\rangle &= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |200\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |21-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |210\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \\ |21+1\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \quad (8.B.5)$$

From this basis and considering the eigenvalues $E_n^{(0)}$ (Eq. (8.B.3)), the unperturbed matrix reads as:

$$\hat{\mathcal{H}}_0 = -E_0 \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 0 & 1/4 \end{pmatrix}. \quad (8.B.6)$$

For the perturbation, let us consider the Hydrogen atom subjected to an electric field of the type:

$$\mathcal{W} = e\mathcal{E}z, \quad (8.B.7)$$

that is, a constant electric field $\mathcal{E}$ applied along the z direction. We now need to construct the perturbation matrix $\hat{\mathcal{W}}$. These calculations are detailed in Box 8.4. We recommend studying this box, especially because it contains information regarding the system's *Selection Rules*. Thus, the per-

turbation in matrix form, using the $\{|nlm\rangle\}$ basis vectors, reads as:

$$\hat{\mathcal{W}} = e\mathcal{E}a_0 \begin{pmatrix} 0 & 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \alpha & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (8.B.8)$$

where $\alpha = \frac{128}{243}\sqrt{2}$ and a_0 is the Bohr radius.

At this point, we have the unperturbed matrix $\hat{\mathcal{H}}_0$ (Eq. (8.B.6)) and the perturbative matrix $\hat{\mathcal{W}}$ (Eq. (8.B.8)). Considering it is a degenerate problem, we start writing the A -reduced matrix:

$$\hat{\mathcal{W}}_A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (8.B.9)$$

and, from Eq. (8.97), it is possible to obtain the first order energy correction for the A domain, as well as the vector $|1a^{(0)}\rangle$. Thus:

$$|1a^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |100\rangle \quad \text{and} \quad E_{1a}^{(1)} = 0. \quad (8.B.10)$$

Similarly, for the B domain, we write the B -reduced matrix as:

$$\hat{\mathcal{W}}_A = e\mathcal{E}a_0 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (8.B.11)$$

where, by using it in Eq. (8.97), we also obtain the first order energy correction for the B domain, as well as the vectors $|nb^{(0)}\rangle$, as follows:

$$|1b^{(0)}\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |21-1\rangle \quad \text{and} \quad E_{1b}^{(1)} = 0 \quad (8.B.12)$$

$$|2b^{(0)}\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |21+1\rangle \quad \text{and} \quad E_{2b}^{(1)} = 0 \quad (8.B.13)$$

$$|3b^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} [|210\rangle - |200\rangle] \quad \text{and} \quad E_{3b}^{(1)} = +3e\mathcal{E}a_0 \quad (8.B.14)$$

$$|4b^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} [|210\rangle + |200\rangle] \quad \text{and} \quad E_{4b}^{(1)} = -3e\mathcal{E}a_0. \quad (8.B.15)$$

Thus, the states $|1b^{(0)}\rangle$ and $|2b^{(0)}\rangle$ do not undergo first order correction due to the perturbation studied. On the other hand, the states $|3b^{(0)}\rangle$ and $|4b^{(0)}\rangle$ show a mixing of the states $|210\rangle$ and $|200\rangle$ – precisely those that satisfy the selection rules (see Box 8.4). Note also that the first order energy correction for the states $|3b^{(0)}\rangle$ and $|4b^{(0)}\rangle$ depends linearly on the applied electric field: this is the *linear* Stark effect. In this development, we will not calculate the vectors $|1a\rangle$ and $|nb\rangle$; however, these vectors should recover $|1a^{(0)}\rangle$ and $|nb^{(0)}\rangle$ in the limit $\mathcal{E} \rightarrow 0$. Finally, a few words about the zero order energy. Similar to what we did in the other examples, we can use Eq. (8.96) to determine this correction,⁵ which reads as:

$$E_{1a}^{(0)} = -E_0 \quad \text{and} \quad E_{1b}^{(0)} = E_{2b}^{(0)} = E_{3b}^{(0)} = E_{4b}^{(0)} = -\frac{E_0}{4}. \quad (8.B.16)$$

Fig. 8.1 summarizes the results obtained up to this point.

Let us now calculate the second order energy correction. However, we know that to obtain such a correction, we need to sum over all states outside the reference domain; but in the case of the Hydrogen atom, the energy levels extend to infinity, making the development complex. To simplify the problem, we will calculate this second order correction only for the state $|1a\rangle$. This correction is given by Eq. (8.130), and for the present case, considering the selection rules and only the *B* degeneracy domain, we have:

$$E_{1a}^{(2)} \approx \frac{|\langle 210 | \hat{\mathcal{W}} | 100 \rangle|^2}{-3E_0/4} = -\frac{4\alpha^2}{3E_0} (e\mathcal{E}a_0)^2, \quad (8.B.17)$$

⁵This zero order energy could have been obtained previously for the $\{|nlm\rangle\}$ basis vector.

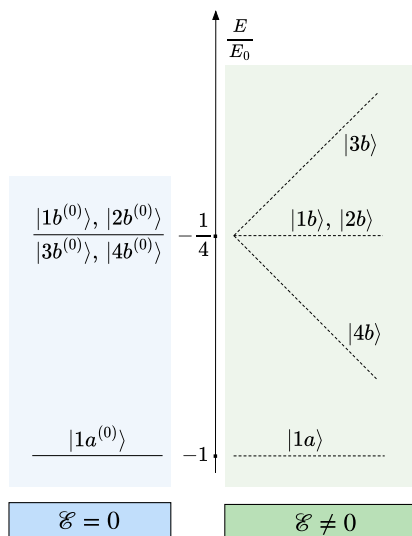


FIGURE 8.1 Energy diagram for the linear Stark effect up to the $n = 2$ shell of the Hydrogen atom. On the right (left), the energy diagram shows the case when the atom is (is not) subjected to an external electric field: perturbed (unperturbed) case. The vectors for the perturbed case were not calculated, only the corresponding energies. However, these vectors should recover $|1a^{(0)}\rangle$ and $|nb^{(0)}\rangle$ in the limit $\mathcal{E} \rightarrow 0$.

where we are considering $|1a^{(0)}\rangle = |100\rangle$. The state $|1a\rangle$ does not have a first order energy correction, but it does have a second order correction. It is easy to see that the second order correction is proportional to $\mathcal{E}^2$, and this term is known as the *quadratic* Stark effect.

★ **Johannes Stark (1874–1957):** Born in Freihung, Germany. In 1919, he received the Nobel Prize in Physics for the discovery of the splitting of spectral lines in electric fields.

Box 8.4 Construction of the perturbation matrix and selection rules

To construct the perturbation matrix, we need to express it using the basis vectors $\{|nlm\rangle\}$, just as we did for the unperturbed term in Eq. (8.B.4). Thus:

$$\begin{aligned} \langle n'l'm' | \hat{\mathcal{V}} | nlm \rangle \\ = e\mathcal{E} \langle n'l'm' | \hat{z} | nlm \rangle \end{aligned}$$

$$= e\mathcal{E} \int_0^\infty \int_0^{2\pi} \int_0^\pi R_{n'l'}^*(r) Y_{l'}^{m'*}(\theta, \phi) r \cos \theta R_{nl}(r) Y_l^m(\theta, \pi) r^2 \sin \theta d\theta d\phi dr \quad (8.B.18)$$

Note that calculating all of the elements can be quite time-consuming, as the wave functions involved are rather complex. For this reason, we need to obtain the *Selection Rules* of the system, which immediately tell us which elements of this matrix are non-zero. However, it is important to note that we need at least two selection rules: one for m and another for l .

Selection Rule for m

For this purpose, let us explore the following commutator (see exercises in Chapter 6):

$$[\hat{L}_z, \hat{z}] = 0 \quad \Rightarrow \quad \langle n'l'm' | [\hat{L}_z, \hat{z}] | nlm \rangle = 0. \quad (8.B.19)$$

Consequently, we find:

$$(m' - m)\hbar \langle n'l'm' | \hat{z} | nlm \rangle = 0. \quad (8.B.20)$$

Above, we have a product of two terms, $(m' - m)\hbar$ and $\langle n'l'm' | \hat{z} | nlm \rangle$, the result of which is zero. To ensure that $\langle n'l'm' | \hat{z} | nlm \rangle$ can be non-zero, we must have $(m' - m)\hbar = 0$, which implies that:

$$m' = m \quad (8.B.21)$$

this being the selection rule for m .

Selection Rule for l

For this purpose, we need to explore the commutator:

$$[\hat{L}^2, [\hat{L}^2, \hat{z}]] = 2\hbar^2 \{\hat{L}^2, \hat{z}\} \quad \Rightarrow \quad \hat{L}^4 \hat{z} - 2\hat{L}^2 \hat{z} \hat{L}^2 + \hat{z} \hat{L}^4 - 2\hbar^2 \hat{L}^2 \hat{z} - 2\hbar^2 \hat{z} \hat{L}^2 = 0 \quad (8.B.22)$$

being $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ the anti-commutator. Just as in Eq. (8.B.19), it is possible to write:

$$\langle n'l'm' | (\hat{L}^4 \hat{z} - 2\hat{L}^2 \hat{z} \hat{L}^2 + \hat{z} \hat{L}^4 - 2\hbar^2 \hat{L}^2 \hat{z} - 2\hbar^2 \hat{z} \hat{L}^2) | nlm \rangle = 0, \quad (8.B.23)$$

and, consequently:

$$\begin{aligned} & \left[l'^2(l'+1)^2 - 2l'(l'+1)l(l+1) + l^2(l+1)^2 - 2l'(l'+1) - 2l(l+1) \right] \\ & \times \hbar^4 \langle n'l'm' | \hat{z} | nlm \rangle = 0. \end{aligned} \quad (8.B.24)$$

Analogously to what we did for the selection rule for m , when the first factor of the multiplication above is zero, the second one *can* be different from zero – and this is the selection rule for l . Therefore, we need to obtain the condition that makes the term in brackets above equal to zero. To this end, we need to reformulate this factor, which contains l' and l , by expanding all the distributive elements and forming a fourth order polynomial in l . After a few calculation steps, which can be easily carried out, we obtain:

$$l^4 + 2l^3 - (1 + 2l' + 2l'^2)l^2 - 2(1 + l' + l'^2)l + (l'^4 + 2l'^3 - l'^2 - 2l') = 0. \quad (8.B.25)$$

This polynomial has four roots^a:

$$\lambda_1 = -2 - l', \quad \lambda_2 = -1 + l', \quad \lambda_3 = 1 + l' \quad \text{and} \quad \lambda_4 = -l', \quad (8.B.26)$$

and then it is possible to rewrite Eq. (8.B.25) as:

$$(l + 2 + l')(l + 1 - l')(l - 1 - l')(l + l') = 0. \quad (8.B.27)$$

The first parenthesis above imposes that $l + 2 + l' = 0$, which is physically impossible, as $(l, l') \geq 0$. The second parenthesis imposes that $l' = l + 1$, while the third parenthesis leads to $l' = l - 1$. Finally, the fourth and last parenthesis implies $l' = -l$ and, consequently, $l' = l = 0$ (the only solution for $l' = -l$). Therefore, we have the following selection rules for l :

$$l' = l \pm 1 \quad \text{and} \quad l' = l = 0. \quad (8.B.28)$$

The objective of the selection rules is to identify the non-zero elements of the perturbative matrix without having to calculate all of its elements. In this way, since we have three possible selection rules for l (Eq. (8.B.28)), we can further simplify the problem by focusing on the selection rule $l' = l = 0 \Rightarrow m' = m = 0$. The matrix element for this selection rule is therefore:

$$\begin{aligned} & \langle n'00 | \hat{\mathcal{V}} | n00 \rangle \\ &= e\mathcal{E} \int_0^\infty \int_0^{2\pi} \int_0^\pi R_{n'0}^*(r) Y_0^{0*}(\theta, \phi) r \cos \theta R_{n0}(r) Y_0^0(\theta, \pi) r^2 \sin \theta d\theta d\phi dr \\ &= \frac{e\mathcal{E}}{4\pi} \int_0^\infty R_{n'0}^*(r) r^3 R_{n0}(r) dr \underbrace{\int_0^{2\pi} d\phi \int_0^\pi \cos \theta \sin \theta d\theta}_{=0} = 0. \end{aligned}$$

This occurs because the $l = 0$ sub-shell is spherically symmetric, and thus the integrand in θ for the above integral only contains $\sin \theta \cos \theta$ – an odd function with a period of π , which, when integrated over the interval $[0, \pi]$, cancels out. Therefore, the valid selection rule for l is:

$$l' = l \pm 1 \quad (8.B.29)$$

Construction of the perturbative matrix

We can now construct the perturbative matrix up to the $n = 2$ shell. For this, we will consider only the states $|100\rangle$, $|200\rangle$, $|21-1\rangle$, $|210\rangle$, and $|21+1\rangle$, written for the $\{|nlm\rangle\}$ basis vector (defined in Eq. (8.B.5)). From the selection rules, we can directly determine which elements are non-zero, namely: $\langle 100|\hat{\mathcal{W}}|210\rangle$ and $\langle 200|\hat{\mathcal{W}}|210\rangle$ (and their respective transposes). Therefore, by solving Eq. (8.B.18) for the above cases, we find^a:

$$\langle 100|\hat{\mathcal{W}}|210\rangle = \alpha e\mathcal{E}a_0 \quad \text{and} \quad \langle 200|\hat{\mathcal{W}}|210\rangle = -3 e\mathcal{E}a_0, \quad (8.B.30)$$

where $\alpha = \frac{128}{243}\sqrt{2}$. Remember that a_0 is the Bohr radius. Thus, the perturbative matrix up to the $n = 2$ shell reads as:

$$\hat{\mathcal{W}} = e\mathcal{E}a_0 \begin{pmatrix} 0 & 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \alpha & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (8.B.31)$$

^aSolution obtained using a software [19].

Appendix 8.A Exercises

Conceptual exercises

Exercise 8.1. The first order correction $|n^{(1)}\rangle$ to the state $|n\rangle$ depends on other states $|m^{(0)}\rangle$. What is the physical meaning of this dependence?

Text-based exercises

Exercise 8.2. Obtain the results of Eq. (8.55).

Exercise 8.3. Obtain a recurrence relation for Eqs. (8.9)–(8.12).

Exercise 8.4. Obtain a recurrence relation for Eqs. (8.14)–(8.17).

Exercise 8.5. Obtain Eq. (8.A.34) for the case $u = y$.

Exercise 8.6. Determine the following commutators:

1. $[\hat{L}^2, \hat{z}]$,
2. $[\hat{L}^2, [\hat{L}^2, \hat{z}]]$.

Practice exercises

Exercise 8.7. Use the perturbative method to obtain the roots of the function $f(x) = x^2 + e^{\epsilon x} - 5$, where $\epsilon \ll 1$. Consider corrections up to second order in ϵ .

Exercise 8.8. Consider (i) the differential equation $\ddot{y}(t) + \epsilon \dot{y}(t) + 1 = 0$, where $\epsilon \ll 1$ and (ii) the initial conditions $y(0) = 0$ and $\dot{y}(0) = 1$. Use the perturbative method to obtain the solution $y(t)$ up to second order in ϵ .

Development exercises

Exercise 8.9. Consider a particle of mass m in a potential well $V(x)$, defined as:

$$V(x) = \begin{cases} 0, & 0 \leq x \leq a, \\ +\infty, & x > a \text{ e } x < 0. \end{cases} \quad (8.A.1)$$

This system is perturbed by a potential $\mathcal{W} = a\omega_0\delta(x - a/2)$. Calculate:

1. The first order energy correction for any state n .
2. The second order energy correction for $n = 1$.

Exercise 8.10. Consider the Zeeman Hamiltonian⁶ $\hat{\mathcal{H}} = \mu_B g \vec{S} \cdot \vec{B}$, where: $\vec{B} = (B_x, 0, B_z)$, $B_z \gg B_x$ and $s = 1/2$. Calculate:

1. The first and second order corrections to the energy and vectors.
2. Solve the exact case. Expand the solution considering $B_z \gg B_x$ and compare with the results from item (1).

Advanced exercises

Exercise 8.11. The dipolar interaction⁷ between two spins s_1 and s_2 can be written as $\hat{\mathcal{H}}_0 = D \hat{S}_{1z} \hat{S}_{2z}$ (there are other terms, but for simplicity, we will ignore them). Consider a magnetic field applied along the x axis and a perturbation⁷ of the form $\hat{\mathcal{W}} = g\mu_B B_x \hat{S}_x$. Obtain the van Vleck susceptibility. Consider $s_1 = s_2 = 1/2$.

Exercise 8.12. Consider the unperturbed Hamiltonian $\hat{\mathcal{H}}_0 = -J \vec{S}_1 \cdot \vec{S}_2$ and a perturbation of the type $\hat{\mathcal{W}} = g\mu_B B_x \hat{S}_x$. Obtain the van Vleck susceptibility, for $s_1 = s_2 = 1/2$.

⁶The angular momentum is dimensionless, as the pre-factor $1/\hbar$ is omitted.

⁷The angular momentum is dimensionless, as the pre-factor $1/\hbar^2$ is omitted.

Exercise 8.13. A particle of mass m in a three-dimensional potential given by:

$$V(x, y, z) = \begin{cases} 0, & 0 \leq (x, y, z) \leq a \\ \infty, & (x, y, z) < 0 \text{ and } (x, y, z) > a \end{cases} \quad (8.A.2)$$

experiences a time-independent perturbation:

$$\mathcal{W} = \frac{a^3}{8} \omega_0 \delta(x - a/2) \delta(y - a/2) \delta(z - a/2). \quad (8.A.3)$$

Consider the sphere $k_x^2 + k_y^2 + k_z^2 \leq 11\pi^2/a^2$, where k_u is the wave vector in the u direction.

1. Write the normalized wave function, the unperturbed energy spectrum and the quantum numbers for this problem.
2. Obtain the selection rules of the system.
3. Construct the perturbation matrix.
4. Obtain the first order energy correction for the considered states.

Exercise 8.14. Consider a Hydrogen atom subjected to a perturbation of the form $\hat{\mathcal{W}} = \epsilon \hat{\mathcal{O}}$, where $\epsilon \ll 1$ and $\hat{\mathcal{O}}$ is an operator such that $[\hat{L}_z^2, \hat{\mathcal{O}}] = [\hat{L}^2, [\hat{L}^2, \hat{\mathcal{O}}]] = 0$.

1. Obtain the selection rules.
2. Construct the perturbative matrix up to $n = 2$. Define the non-zero elements as a, b, c , etc., and use the smallest possible number of these parameters.

Exercise 8.15. Consider two spins $s_1 = s_2 = 1/2$ interacting through the Heisenberg Hamiltonian $\hat{\mathcal{H}}_0 = -J \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2$ and a perturbation of the type $\hat{\mathcal{W}} = \frac{\lambda}{2} (\hat{S}_+^2 + \hat{S}_-^2)$, where $\vec{\hat{S}} = \vec{\hat{S}}_1 + \vec{\hat{S}}_2$.

1. Derive the selection rules for the system.
2. Using the results from the previous item, construct the perturbative matrix, identifying the non-zero elements as $a, b, c, \dots$
3. Calculate the elements of the perturbative matrix.
4. Create a table listing the unperturbed energy of each state, as well as the first and second order energy corrections.